

4/21/26

EXAM 4: FORM D

Name:

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UCF ID:

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CHM2045C SPRING 2026

1. Write your name on your scratch paper.

I will keep your bubble Sheet. You can keep your scratch paper. I recommend writing your answers on your scratch paper so you can compare them to the key when I post it. You can view your test after I post the key on Webcourses within a week of the exam. The highest score on the exam is 100 points.

2. Write your **name** and **UCFID** on the bubble sheet and exam. **Bubble in your test form.**
3. When I give the signal to begin, you may open your test and begin working. This is an individual assessment – you may not work with anyone else. The reference sheet provided and your non-programmable calculator are the only approved resources – you may not use other notes, books, electronics, etc.
4. If you are caught with any electronic device, regardless of if you are using it. You will receive a zero (0).
5. When you are finished, show your ID and give your exam and bubble sheet to a proctor. I will collect all remaining exams after 50 minutes.
6. Any student not enrolled in CHM2045C section 0004 taking the exam will be reported. Any student who copies, distributes, and/or removes the exam will receive a **Z grade for academic misconduct**.
7. When you are finished, show your ID and give your exam and bubble sheet to a proctor. You must wait 25 minutes to turn in the exam. I will collect all remaining exams after 50 minutes.

Fundamental Physical Constants

Avogadro's Number	$N_A = 6.02214 \times 10^{23} / \text{mol}$
Atomic Mass Unit	$\text{amu} = 1.66054 \times 10^{-27} \text{ kg}$
Charge of the Electron	$e = 1.60218 \times 10^{-19} \text{ C}$
Faraday Constant	$F = 9.64853 \times 10^4 \text{ C/mol}$
Mass of the Electron	$m_e = 9.10939 \times 10^{-31} \text{ kg}$
Mass of the Neutron	$m_n = 1.67493 \times 10^{-27} \text{ kg}$
Mass of the Proton	$m_p = 1.67262 \times 10^{-27} \text{ kg}$
Planck's Constant	$h = 6.62607 \times 10^{-34} \text{ J}\cdot\text{s}$
Speed of Light	$c = 2.99792 \times 10^8 \text{ m/s}$
Acceleration of Gravity	$g = 9.80665 \text{ m/s}^2$
Rydberg Constant	$R_H = 1.09677 \times 10^7 \text{ m}^{-1}$
Universal Gas Constant	$R = 8.31447 \text{ J/mol}\cdot\text{K}$ $R = 0.082058 \text{ L}\cdot\text{atm/mol}\cdot\text{K}$

Conversions and Relationships

Length	1 km = 1×10^3 m = 0.621 mile 1 inch = 2.54 cm 1 ft = 12 in 1 pm = 1×10^{-12} m = 0.01 Å
Mass	1 kg = 1×10^3 g = 2.205 lb 1 metric ton = 1×10^3 kg
Volume	1 dm ³ = 1×10^{-3} m ³ = 1 liter 1 cm ³ = 1 mL 1 m ³ = 35.3 ft ³ 1 gallon = 3.785 liters
Energy	1 J = 1 kg·m ² /s ² = 1 C·V 1 calorie = 4.184 J
Temperature	$T(\text{K}) = T(^{\circ}\text{C}) + 273.15$ $T(^{\circ}\text{C}) = (T(^{\circ}\text{F}) - 32)(5/9)$ H ₂ O: mp = 0°C; bp = 100°C
Pressure	1 Pa = 1 N/m ² = 1 kg/m·s ² 1 atm = 1.01325×10^5 Pa 1 atm = 760 torr = 760 mmHg
Math	$\pi = 3.1416$ e = 2.7183

180, ~~120~~, 90/120, 90, Planes, Tetra, Tripura, Oct
120, ~~109~~, 109 Bent Biaya Bent
C. Venkay Owen Lopez

Name _____

Name

FORM D

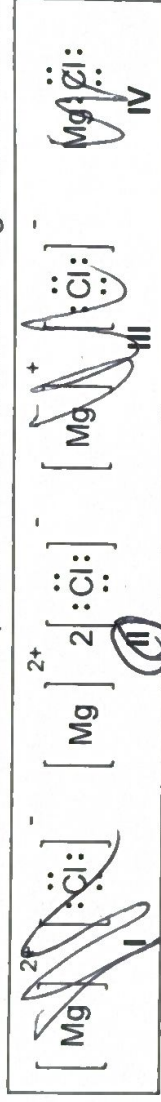
MULTIPLE CHOICE. Choose the answer(s) that best answers the question. Circle your answer on the exam and bubble your choice on your Bubble Sheet.

1. Which of the following occurs as the wavelength of a photon increases?
 A) The frequency decreases.
 B) The energy increases.
 C) The frequency decreases.
 D) The frequency increases.
 E) 200°
2. Give the approximate bond angle for a molecule with a tetrahedral shape.
 A) 105°
 B) 30°
 C) 120°
 D) 109.5°
 E) 200°
3. Correctly assign a formal charge to the nitrogen atom of the thiocyanate ion (SCN)⁻ shown below.

$$:\text{S} \equiv \text{C} - \text{N}:: 7-7$$

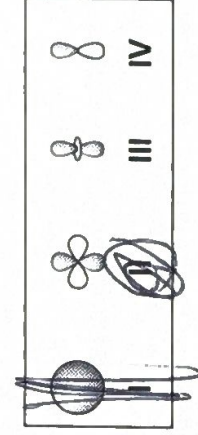
- A) +1 B) -1 C) 0 D) -2

4. What is the Lewis structure for the ionic compound formed between magnesium and chlorine?



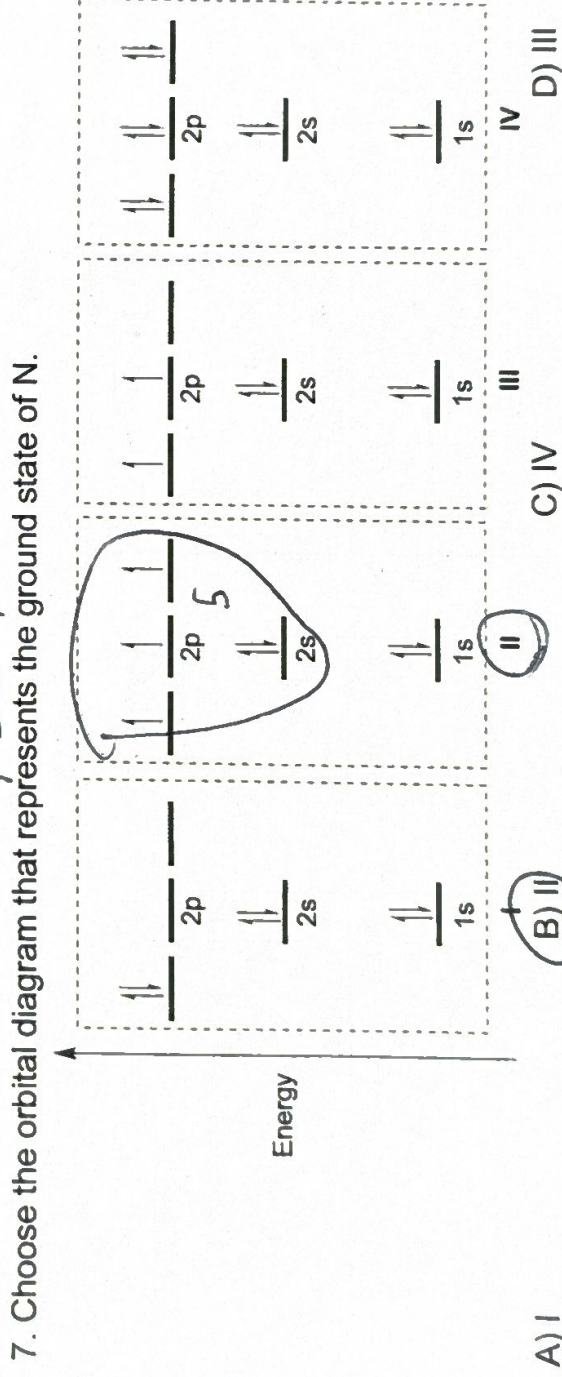
- A)  B) IV
5. Which best represents the shape of a $3p_y$ orbital?

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6. What are the possible orbitals for $n = 4$?
- A) s, p, d, f B) s, p C) s, p, d, f D) s

7. Choose the orbital diagram that represents the ground state of N.



8. Determine which pair of atoms produces a polar covalent bond.

Atom	Electronegativity	Atom	Electronegativity
C	2.5	Cl	3.0
P	2.1	S	2.5
O	3.5	H	2.1

[illegible]

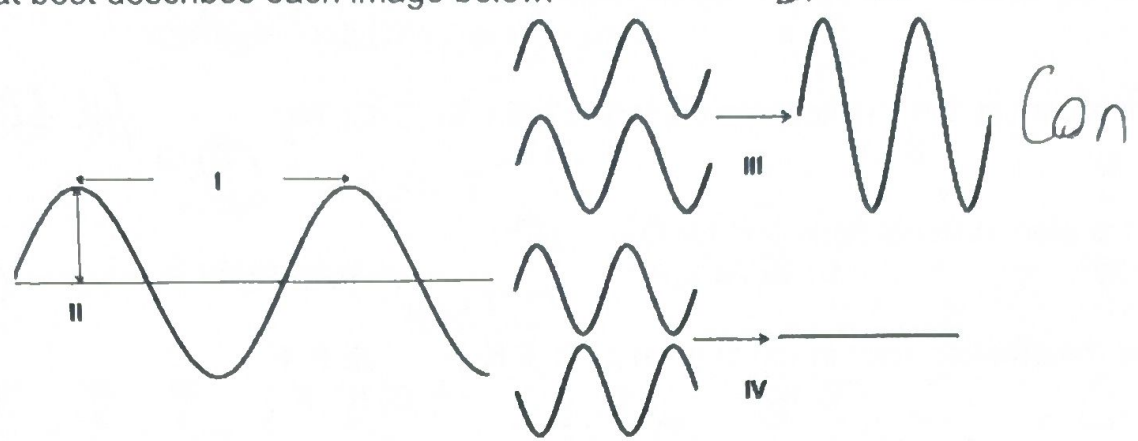
- A) I and IV
B) IV and V
C) II and III
D) II, III, and V
E) III and V

I, II, IV

$$\begin{array}{r} 2.6-3.0 = 0.5 \text{ PL} \\ 2.5-3.5 = 1 \text{ PC} \\ 2.1-2.1 = 3.6-2.1 \end{array}$$

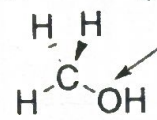
I: Freq
II: Wavelength
III: Con
IV: Destruct

9. Select the terms that best describes each image below:



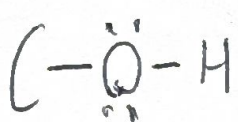
- (A) I = frequency; II = wavelength; III = constructive interference; IV = destructive interference
 B) I = wavelength; II = amplitude; III = constructive interference; IV = destructive interference
~~C) I = amplitude; II = wavelength; III = constructive interference; IV = destructive interference~~
~~D) I = amplitude; II = wavelength; III = positive interference; IV = negative interference~~

10. Determine the electron geometry (eg) and molecular geometry (mg) of the atom indicated in the molecule.



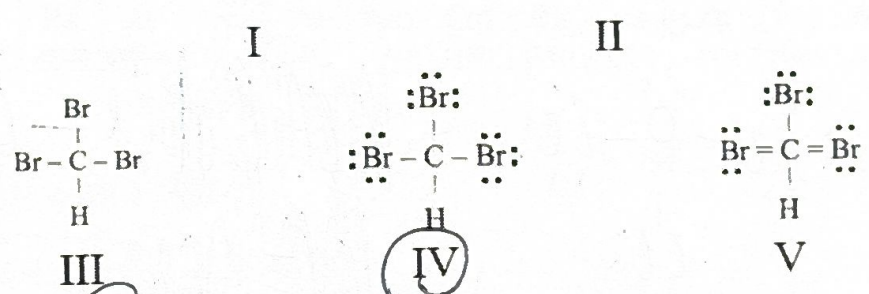
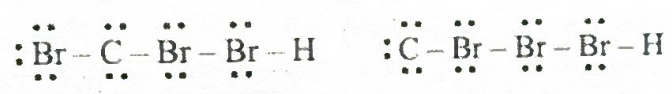
4, Tetrahedral

- ~~A) eg = bent; mg = tetrahedral~~
 B) eg = tetrahedral; mg = linear
 C) eg = tetrahedral; mg = bent
~~D) eg = trigonal bipyramidal; mg = tetrahedral~~



4 2 tetrahedral Bent

11. Choose the best Lewis structure for CHBr3.



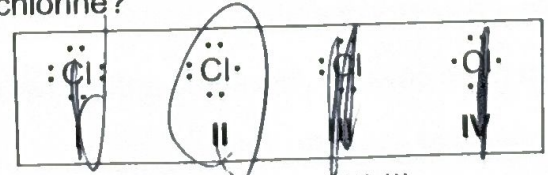
- A) II
 B) IV
 C) III
 D) I

12. Of the following, which element has the **lowest** first ionization energy? $IE = 4R$
 A) Ge B) Se C) Br D) Rb

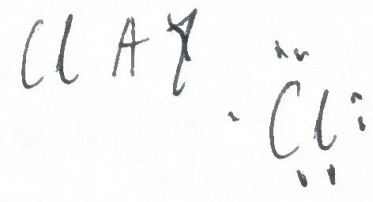
13. Determine the energy change associated with the transition from $n=3$ to $n=2$ in the hydrogen atom.
 A) $-3.03 \times 10^{-19} \text{ J}$ B) $-1.82 \times 10^{-19} \text{ J}$ C) $+3.03 \times 10^{-19} \text{ J}$ D) $+5.51 \times 10^{-19} \text{ J}$

$\Delta E = (-2.18 \times 10^{-18}) \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$
 $= (-2.18 \times 10^{-18}) \left(\frac{1}{4} - \frac{1}{9} \right) = (-2.18 \times 10^{-18}) \left(\frac{9-4}{36} \right) = (-2.18 \times 10^{-18}) \left(\frac{5}{36} \right) = -3.03 \times 10^{-19} \text{ J}$
 Energy Release from system (-)

14. What is the Lewis symbol for chlorine?



- A) IV
 B) I
 C) III
 D) I



15. B

15. How many valence electrons does an atom of sulfur have?

A) 1

B) 6

C) 4

D) 3

16. D

16. Which atom has the smallest atomic radius? Na, Si, S, Cl, Mg

A) S

B) Si

C) Na

D) Cl

E) Mg

17. D

17. Give the electronic configuration for O.

A) $1s^2 2s^2 2p^3$

B) $1s^2 2s^2 2p^5$

C) $1s^2 2s^2 2p^6$

D) $1s^2 2s^2 2p^4$

18. D

18. Chose the **smaller** atom or ion in each pair: I: K, K^+ II: F, F^-

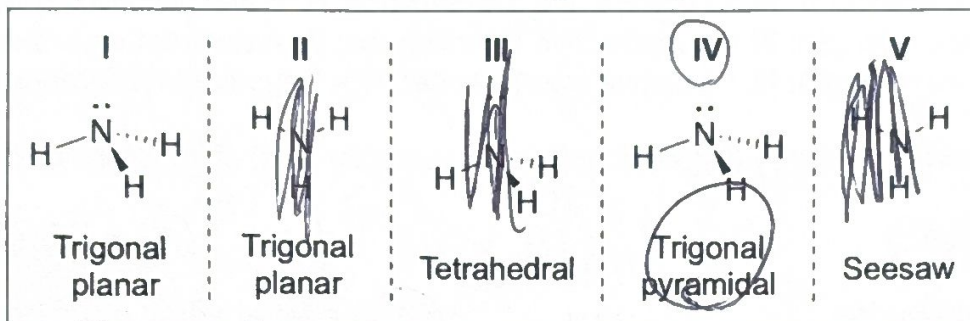
A) K, F^-

B) K, F

C) K^+ , F^-

D) K^+ , F

19. Which of the following best represents the molecular geometry of NH_3 ?



A) III

B) I

C) V

D) II

E) IV

20. Use the bond energies provided to estimate ΔH°_{rxn} for the reaction below.



$\Delta H^\circ_{rxn} = ?$

Bond

Cl-Cl

P-Cl

Bond Energy (kJ/mol)

243

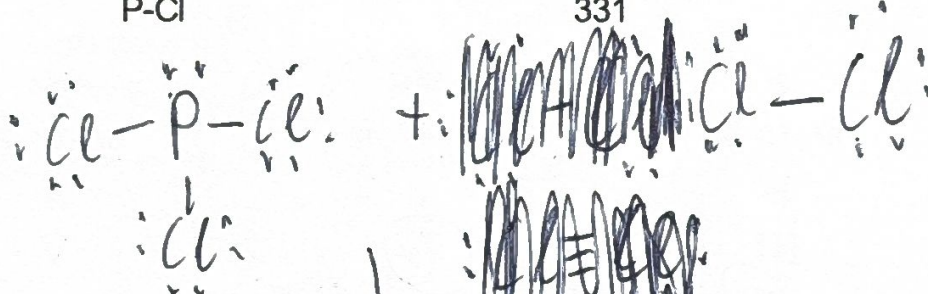
331

A) -243 kJ

B) -67 kJ

C) -419 kJ

D) -662 kJ

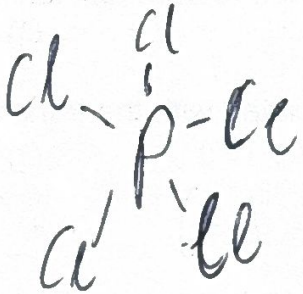


$$3 \text{ Cl-P} - 993$$

$$1 \text{ Cl-Cl} - 243$$

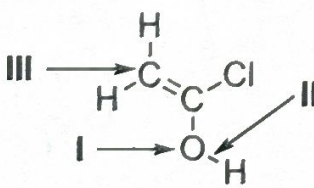
$$+ 5 \text{ Cl-P} \quad 1655$$

$$419$$



21. FR Bonus. **Only answers written on the back of the Bubble Sheet will be graded.**

Using the molecule below, answer the following questions.



a) Molecular geometry of atom I:

b) Bond in II (pure covalent, covalent, polar covalent, ionic):

c) Formal charge of atom III:

21.

1A
1

8A
18

Actinides

Ge Se Br Rb

$$\Delta H = \Delta E + \Delta(PV)$$

$$q = mc \Delta T$$

$$w = -P_{ext}\Delta V$$

$$\Delta H_{rxn}^\circ = \sum \text{mol} \cdot \Delta H_f^\circ(\text{products}) - \sum \text{mol} \cdot \Delta H_f^\circ(\text{reactants})$$

$$\Delta H_{rxn}^{\circ} = \sum \text{mol} \cdot BE (\text{bonds broken}) - \sum \text{mol} \cdot BE (\text{bonds formed})$$

$$\lambda = \frac{h}{mv} \quad c = \lambda \nu \quad \Delta E = h\nu \quad \Delta E = \frac{hc}{\lambda}$$

$$\Delta E = -2.18 \times 10^{-18} \text{ J} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$M = \text{molar mass} \quad d = \frac{MP}{RT} \quad M = \frac{mRT}{PV} \quad PV = nRT$$

$$\frac{P_1 V_1}{n_1 T_1} = \frac{P_2 V_2}{n_2 T_2} \quad KE = \frac{3}{2} RT = \frac{1}{2} m v^2$$

$$F_e = \frac{kQ_1Q_2}{d^2}$$

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\frac{\text{Rate}_A}{\text{Rate}_B} = \frac{\sqrt{M_B}}{\sqrt{M_A}}$$

$$P_A = X_A \cdot P_{\text{total}}$$

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

$$\ln\left(\frac{P_2}{P_1}\right) = \frac{-\Delta H_{vap}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right) \quad \ln\left(\frac{k_2}{k_1}\right) = \frac{-E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$k = Ae^{-E_a/RT}$$

$$P_{\text{solvent}} = X_{\text{solvent}} P_{\text{solvent}}^{\circ} \quad \Delta P = (X_{\text{solute}} P_{\text{solvent}}^{\circ}) i$$

$$Na \quad Si \quad S \quad \underline{Cl} \quad Mg$$

SCRATCH PAPER

91 29 03 20

27 22 26